higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

ENGINEERING FABRICATION
(First Paper)
NQF LEVEL 2

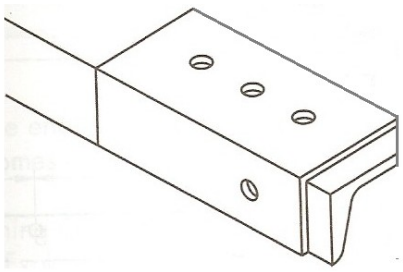
22 NOVEMBER 2019

This marking guideline consists of 5 pages.

Copyright reserved

QUESTION 1

1.1



(4)

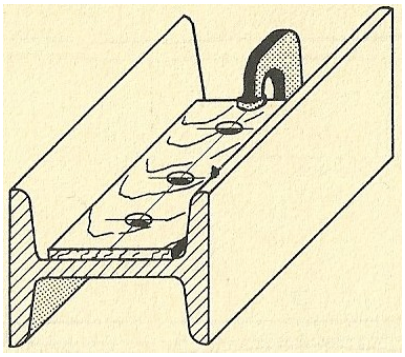
1.2
- Wood
- Cardboard
- Thin sheet of plate
- Paper

(4)

1.3 Parallel line method (1)

1.4 Back mark is the distance measured from the heel of an angle iron to the centre of the hole on the leg of an angle iron, while the back-mark gauge is a tool that marks off back marks on the leg of an angle iron. (3)

1.5



(3)
[15]

QUESTION 2

2.1 Back mark gauge is a tool to mark off back marks on the leg of an angle iron whereas bevel gauge is a tool to measure or transfer angles. (3)

2.2 2.2.1 V = A × H✓ = 3,142(R × R)H✓
= 3,142 (0,6 × 0,6) × 1,8✓
= 2,036 m cubed✓

2.2.2 A = C × H✓
= (MD × 3,142) × H✓
= (1,2 + 0,016) × 3,142 × 1,8✓
= 6,877 m squared✓

(2 × 4) (8)

2.2.3 Malleability and ductility (2)

2.3 2.3.1 True
2.3. 2 False

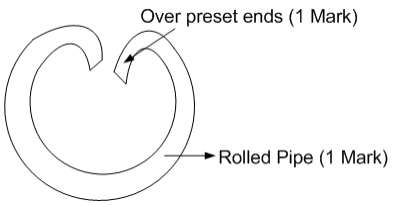
(2 × 1) (2)
[15]

QUESTION 3

3.1
- The Punch
- The Shearer
- The Cropper
- Bending and Rolling machines (Any 3 × 1) (3)

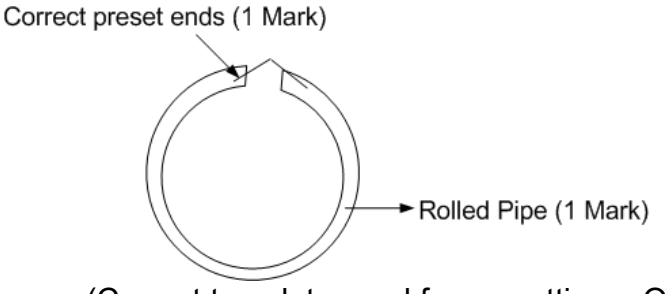
3.2 Bending machines are mainly used to bend sheet metal plates whereas forming machines are mainly used for rolling cylinders. (4)

3.3 3.3.1



(NO template used during presetting = ONE mark) (3)

3.3.2



(Correct template used for presetting = ONE mark) (3)

3.4
- Loose clothing
- Long hair that is not tied up
- Jewellery (rings and necklaces)
- NO safety glasses

(Any 2 × 1) (2)

[15]

QUESTION 4

4.1 4.1.1 Oxygen regulator (1)

4.1.2
- It reads the amount of content in the cylinder.
- It regulates the amount of gas flow from the cylinder. (2)

4.2 Neutral flame (1)

4.3 4.3.1 Backfiring is a feeding back of flame into the cylinder whereas flashback is equipment that is used to prevent backfiring. (3)

4.3.2 Carburising flame: has too much of carbon.

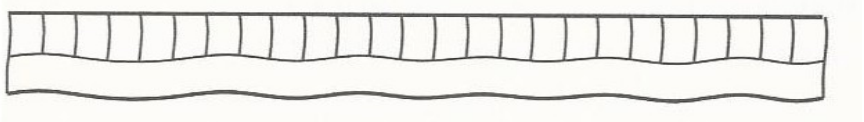


Neutral flame: equal quantities of oxygen and acetylene.

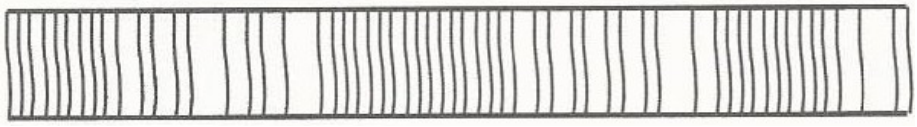


(2 × 2) (4)

4.4.1 Nozzle is held too high from the metal being cut.



4.4.2 Nozzle is held too close to the metal being cut



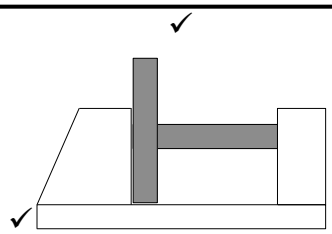
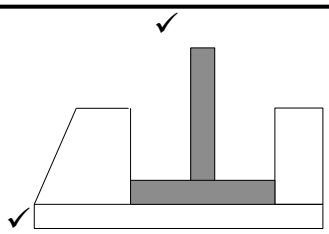
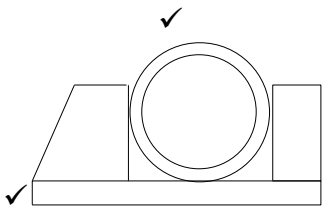
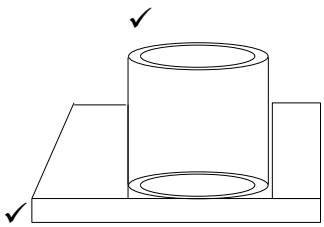
(2 × 2) (4)

[15]

QUESTION 5

5.1
- Reduce friction between sliding surfaces
- Prevent ware on machines
- Cooling the moving parts
- Prevent corrosion
- Prevent things sticking together (adhesion) (Any 3 × 1) (3)

5.2

STEEL SECTION	CORRECT CLAMPING	INCORRECT CLAMPING
T-bar		
Round pipe		

(2 × 4) (8)

5.3 In dry cutting, a cutting disc is used without a coolant whereas is wet cutting a coolant is used to cool the blade. (4)

[15]

TOTAL: 75

Copyright reserved